Department of Computer Engineering
CMPE 491 — Senior Design Project I

Project Analysis Report

TEDU Guide

AI-Powered Intelligent Campus Navigation System

Project Website: https://tedu-guide.vercel.app/

Team Members

1	Taci Furkan Gökalp
2	Elif Nazlı Özer
3	Fatima Nasir Muhammad

Advisor & Jury Members

Advisor	Tolga Kurtuluş Çapın
Jury Member 1	Mehmet Evren Coşkun
Jury Member 2	Müslüm Bozyiğit

Spring 2025–2026 · Submitted: April 2026

Table of Contents

1. Introduction

    1.1 Purpose and Scope
    1.2 Document Structure
2. Current System

    2.1 Existing Navigation Mechanisms
        2.1.1 Static Physical Signage
        2.1.2 Printed Floor Plans
        2.1.3 Manual Inquiry
    2.2 Limitations of the Current System
3. Proposed System

    3.1 Overview
    3.2 Functional Requirements
    3.3 Non-Functional Requirements
    3.4 Pseudo Requirements
    3.5 System Models
        3.5.1 Scenarios
        3.5.2 Use Case Model
        3.5.3 Object and Class Model
        3.5.4 Dynamic Models
        3.5.5 User Interface
    3.6 Risk Analysis and Mitigation Strategies
    3.7 System Validation and Evaluation Plan
    3.8 Error Handling and Edge Cases
4. Glossary

5. References

1. Introduction

This document makes up the Project Analysis Report (D4) for TEDU Guide, an AI-powered intelligent campus navigation system developed for TED University. The purpose of this report is to provide a comprehensive analysis of the proposed system, establishing a contract between the development team and stakeholders that defines precisely what the system will do, how users will interact with it, and how the key components will behave under various conditions.

The analysis report serves as the bridge between the high-level specifications defined in D2 (Project Specification Report) and the forthcoming system design. It introduces detailed models, including use case scenarios, object class models, dynamic models, and user interface prototypes to ensure all stakeholders share a common, unambiguous understanding of the system before implementation begins.

1.1 Purpose and Scope

The TEDU Guide addresses the challenges of indoor navigation within TED University's multi-floor campus buildings. The system aims to provide students, faculty, staff, and campus visitors with an intuitive, AI-driven interface for locating rooms, offices, laboratories, and campus facilities in real time.

The scope of this analysis includes the complete software system, including the user-facing mobile-compatible web application, the backend API server, the WiFi-based indoor positioning subsystem, the NLP query interface powered by OpenAI GPT-4, the interactive floor plan rendering layer, and the campus information database. The system is bounded to TED University's campus buildings, where WiFi fingerprinting data will be collected.

1.2 Document Structure

This report is organized as follows:

- Section 2: Current System: Describes the existing manual navigation methods and their limitations.
- Section 3: Proposed System: Presents the full analysis of the TEDU Guide system, including functional requirements, non-functional requirements, pseudo requirements, and all system models.
- Section 4: Glossary: Defines key technical terms used throughout the document.
- Section 5: References: Lists all sources cited.

2. Current System

TED University currently does not provide any digital indoor navigation system. Campus navigation relies solely on informal methods(asking people for directions) and manual maps and signs, that consistently prove inadequate, particularly during high-traffic periods such as the first weeks of each semester, examination periods, and official open days.

2.1 Existing Navigation Mechanisms

2.1.1 Static Physical Signage

Corridor signage and room number plates are the primary wayfinding tools currently in use. While these serve basic orientation needs, they provide no route guidance, no real-time information, and are frequently insufficient for visitors who are unfamiliar with the building layout. Signage is not consistently placed at decision points (stairwells, corridor intersections), leading to confusion, especially in multi-floor buildings.

2.1.2 Printed Floor Plans

Static printed floor plan maps are available in some building lobbies and on the university intranet. However, these plans are rarely updated when room assignments change, do not indicate the user's current position, and cannot provide route guidance. They require the user to mentally map their path without any real-time assistance.

2.1.3 Manual Inquiry

The most common navigation strategy for first-time visitors and new students is to ask other people—security personnel, other students, or reception staff—for directions. This approach is time-consuming, inconsistent in accuracy, and unavailable outside operational hours.

2.2 Limitations of the Current System

Limitation	Impact
No real-time positioning	**Users cannot determine their current location automatically inside buildings**
Static information	**Floor plans and signage do not reflect current room assignments or facility hours.**
No route guidance	**No turn-by-turn or visual route assistance exists for multi-floor navigation.**
Accessibility gap	**No dedicated wayfinding support exists for users with mobility or visual impairments.**
No NLP interface	**Users cannot submit natural language queries; no intelligent query processing is available.**
Inconsistent experience	**Navigation quality depends entirely on human availability and local knowledge.**

3. Proposed System

3.1 Overview

TEDU Guide is a mobile-compatible web application that integrates three core technologies to deliver a seamless indoor navigation experience: WiFi fingerprinting for real-time indoor positioning, a GPT-4-powered NLP engine for natural-language query processing, and an interactive Leaflet.js floor-plan viewer for route visualization.

The system is designed to be accessible via any modern browser without installation, ensuring a low barrier to entry for all user groups. The technical stack comprises a React.js frontend, a Python FastAPI backend, and a PostgreSQL/PostGIS spatial database. The system is deployed on cloud-hosted infrastructure (Vercel for the frontend, Render/Railway for the backend) and is publicly accessible at https://tedu-guide.vercel.app/.

Figure 6: System Architecture Overview

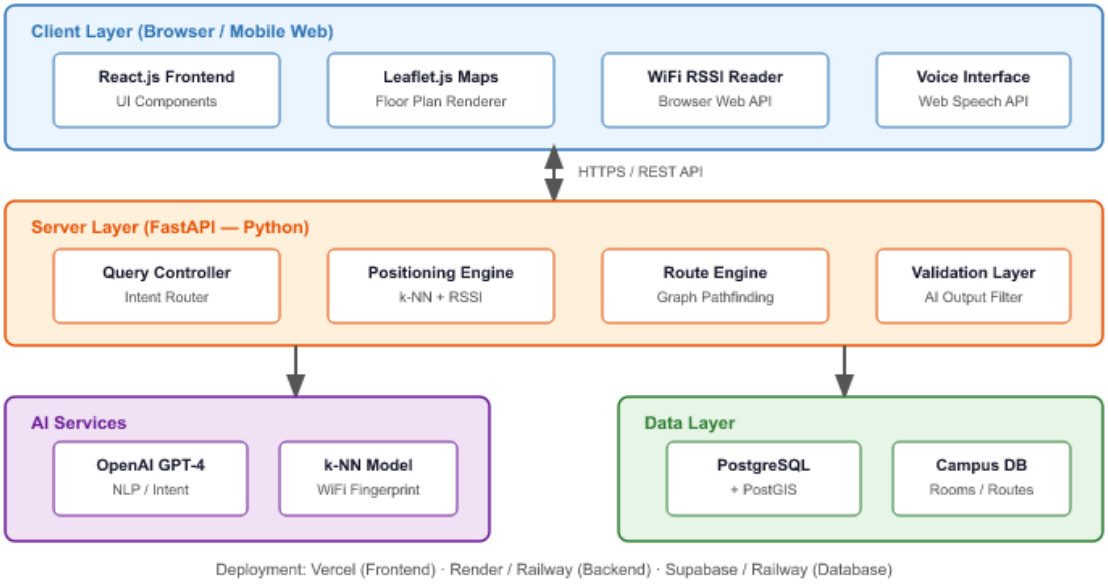

Figure 6: System Architecture Overview

The system architecture is organized into three distinct layers: the Client Layer (React.js frontend, Leaflet.js maps, WiFi RSSI reader, voice interface), the Server Layer (FastAPI backend handling query routing, positioning, route computation, and AI output validation), and the Data Layer (PostgreSQL/PostGIS campus database alongside the OpenAI GPT-4 API and the local k-NN positioning model). All communication between the client and server is encrypted via HTTPS.

3.2 Functional Requirements

The following table presents the functional requirements of the TEDU Guide system. Each requirement uses the 'shall' convention to indicate mandatory behavior, and 'should' for optional enhancements.

ID	Name	Description
FR-01	**Indoor Positioning**	The system shall determine the user's real-time indoor location by collecting and analyzing WiFi RSSI signals from nearby campus access points using a pre-trained k-NN fingerprinting model.

ID	Name	Description
FR-02	**Location Accuracy**	The indoor positioning subsystem shall localize the user's position to within 3–5 meters under standard campus WiFi signal conditions in calibrated building areas.
FR-03	**NLP Query Processing**	The system shall accept free-form natural language queries (text or voice) and parse the user's intent to return relevant campus information, including room locations, office assignments, facility hours, and navigation routes.
FR-04	**Route Generation**	The system shall compute and display the optimal indoor route between any two points on campus, accounting for floor transitions via staircases or elevators, rendered as a highlighted path on the interactive floor plan.
FR-05	**Accessibility Routing**	The system shall provide an accessibility-aware navigation mode that restricts route options to step-free paths and elevator corridors, activated when the user enables the accessibility option.
FR-06	**Interactive Floor Plan**	The system shall render interactive, zoomable floor plan maps for each campus building, displaying rooms, corridors, staircases, elevators, and labeled points of interest.
FR-07	**Campus DB Query**	The system shall maintain a structured database of campus locations—including room numbers, office assignments, faculty names, facility types, and operating hours—which is queried in response to user requests.
FR-08	**Admin Interface**	Authorized administrators shall be able to add, edit, and remove location entries, update room assignments, and modify facility operating hours through a secured administrative web interface.
FR-09	**Voice-Only Mode**	The system shall support a voice-only interaction mode in which all navigation instructions and location information are delivered via synthesized audio output.
FR-10	**Offline Capability**	The system shall remain functional for 30–90 seconds of network disconnection, serving cached navigation data and the most recently loaded floor plan.
FR-11	**Haptic Feedback (Optional)**	When operating on a mobile device, the system should provide haptic vibration feedback to indicate turn directions: one pulse for left, two for right.
FR-12	**Guest Access**	The system shall be accessible to campus visitors connecting via the university's guest WiFi network, without requiring authentication or university account credentials.

3.3 Non-Functional Requirements

ID	Name	Description
NFR-01	**Response Time**	The system shall respond to navigation and information queries within 2 seconds under normal network conditions and typical server load.
NFR-02	**Scalability**	The system shall support concurrent usage by at least 100 simultaneous users without measurable degradation in response time or positioning accuracy.
NFR-03	**Availability**	The system shall target 99% uptime during university operational hours (08:00–22:00 on weekdays).
NFR-04	**Usability**	The interface shall be fully functional on desktop and mobile web browsers without native app installation, operable by a first-time user without prior training.
NFR-05	**Security**	All client–server communication shall be encrypted via HTTPS/TLS. User location data shall not be persistently stored on the server beyond the active session without explicit user consent.
NFR-06	**Privacy**	The system shall comply with KVKK (Law No. 6698) and implement data minimization principles, retaining only session-scoped location data unless the user explicitly opts in to extended storage.
NFR-07	**Maintainability**	The codebase shall be organized into modular components with documented interfaces, enabling independent updates to the positioning model, NLP integration, floor plan data, and campus database.
NFR-08	**Portability**	The system shall be deployable on standard cloud hosting platforms and shall not depend on proprietary infrastructure specific to TED University's internal systems.

3.4 Pseudo Requirements

ID	Name	Description
PR-01	**Technology Stack**	The frontend shall be implemented using React.js with Leaflet.js. The backend API shall use Python with FastAPI. Spatial location data shall be stored in PostgreSQL with the PostGIS extension.
PR-02	**AI/NLP Engine**	The NLP component shall be implemented using the OpenAI API (GPT-4 or equivalent) with prompt engineering to constrain responses to campus-relevant information.
PR-03	**Positioning Algorithm**	The indoor positioning component shall implement WiFi fingerprinting using RSSI data from campus access points, trained with a k-Nearest Neighbors supervised model.
PR-04	**Deployment Platform**	The application shall be deployed on publicly accessible cloud hosting (Vercel for frontend, Render or Railway for backend) with a custom domain linked to the project website.

3.5 System Models

3.5.1 Scenarios

Scenarios describe realistic usage episodes that the system must support. Each scenario is presented in narrative form and maps to one or more functional requirements.

Scenario 1 — New Student Locating a Classroom (Primary Use Case)

Actor	First-year student, Arda, is attending his first lecture
Precondition	User opens TEDU Guide in browser; device connected to campus WiFi
Main Flow	1. Arda types "Where is classroom A-204?" in the search bar. 2. The NLP engine parses his query, extracting intent=navigate and entity=A-204. 3. The system reads WiFi RSSI values from nearby access points and estimates Arda's location as Corridor B, Floor 1. 4. The route engine computes the shortest path from Arda's current position to A-204, including a staircase ascent to Floor 2. 5. The interactive map highlights the route in red. Turn-by-turn text instructions appear in the navigation panel. 6. As Arda walks, his position updates on the map in real time. He receives a haptic vibration cue at each turn. 7. Upon arriving at A-204, the system displays an "Arrived" confirmation and offers a new search.
Postcondition	The user has successfully navigated to the destination. Session ends, or new query begins.
Requirements	FR-01, FR-02, FR-03, FR-04, FR-06, FR-11; NFR-01, NFR-04

Scenario 2 — Visitor Finding a Faculty Office (Secondary Use Case)

Actor	External visitor attending a meeting with Prof. Çapın
Main Flow	1. Visitor connects to TEDU guest WiFi and opens tedu-guide.vercel.app. 2. She types "Prof. Çapın's office" — no account login required. 3. NLP extracts intent=locate, entity=Prof. Çapın. DB returns to the office room 304, Block A. 4. Map highlights Room 304 and provides a route from the main entrance.
Requirements	FR-03, FR-04, FR-06, FR-07, FR-12; NFR-04

Scenario 3 — User with Mobility Impairment Requests Accessible Route

Zeynep is a wheelchair user attending a lecture on Floor 3. She enables Accessibility Mode in TEDU Guide before submitting her query. The system restricts route computation to step-free paths, directing her exclusively through elevator corridors and accessible entrances, and explicitly marks inaccessible paths as unavailable. The turn-by-turn voice output is activated automatically in accessibility mode, reducing the need for screen interaction while navigating. (Covers: FR-05, FR-09, NFR-04)

Scenario 4 — Administrator Updates Room Assignments

A department coordinator logs into the TEDU Guide admin interface using university credentials. She navigates to the Campus Database management panel and changes the

room assignment for Prof. Bozyiğit's office from Room 210 to Room 215 following an office relocation. The change is immediately reflected in subsequent user queries without any code-level modification or system restart. (Covers: FR-07, FR-08; NFR-07)

3.5.2 Use Case Model

The use case model identifies all external actors that interact with the TEDU Guide system and the use cases each actor participates in. The system boundary encloses all functions provided by TEDU Guide; actors reside outside this boundary.

Actors

Actor	Description
Student / Faculty	**Primary end users who use the system to navigate campus buildings and locate offices, classrooms, and facilities.**
Campus Visitor	**External users who access the system via guest WiFi without university credentials. Limited to navigation and information queries.**
Administrator	**Authorized university staff who manage the campus database, update room assignments, and recalibrate the positioning model.**

Figure 1: TEDU Guide — Use Case Diagram

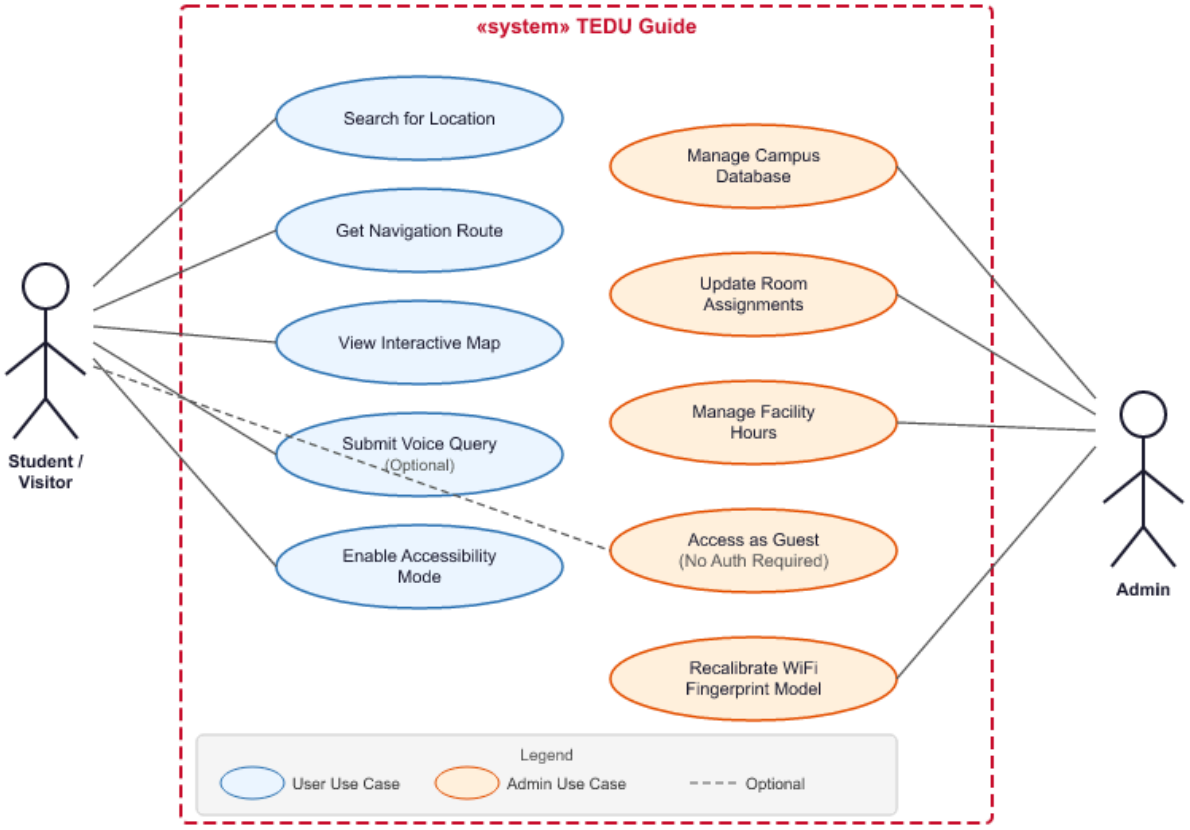

Figure 1: TEDU Guide — Use Case Diagram

Use Case Descriptions

UC-01: Search for Location — The user enters a natural language query (e.g., "Where is the cafeteria?"). The system invokes the NLP engine to parse intent and entities, queries the campus database, and returns a list of matching locations with room numbers and building information.

UC-02: Get Navigation Route — Following a location search, the user selects a result and requests a route. The system determines the user's current position via WiFi fingerprinting, computes the shortest path using graph-based pathfinding, and renders the route on the interactive floor plan.

UC-03: View Interactive Map — The user browses the floor plan of any campus building, zooms in/out, and explores rooms and points of interest. No query is required; the map can be browsed freely.

UC-04: Submit Voice Query — The user activates the voice input mode and speaks a navigation query. The system transcribes the speech using the Web Speech API and processes it identically to a text query.

UC-05: Enable Accessibility Mode — The user toggles the accessibility option, restricting all subsequent route computations to step-free, elevator-accessible paths.

UC-06: Manage Campus Database (Admin) — An authenticated administrator adds, edits, or removes location records via the admin interface. Changes propagate immediately to user-facing queries.

UC-07: Recalibrate WiFi Model (Admin) — An administrator triggers a recalibration workflow to retrain the k-NN model after changes to the campus WiFi access point layout.

3.5.3 Object and Class Model

The class model identifies the key domain objects in the TEDU Guide system, their attributes, responsibilities, and inter-relationships. The diagram uses a simplified UML notation with stereotypes indicating abstract classes and component types.

Figure 2: TEDU Guide — Object and Class Model

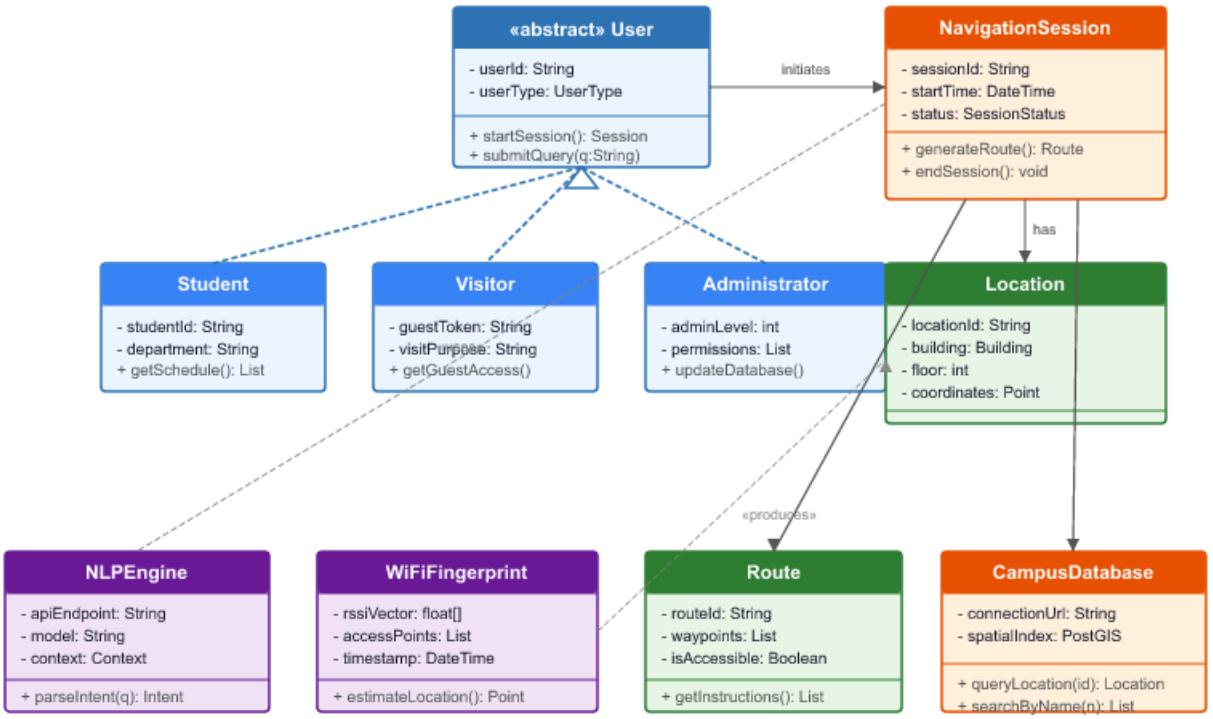

Figure 2: TEDU Guide — Object and Class Model

The User abstract class is the root of the user hierarchy, extended by Student, Visitor, and Administrator. A NavigationSession is initiated by a User and aggregates the core processing components: NLPEngine (responsible for intent parsing via the OpenAI API), WiFiFingerprint (RSSI measurement and k-NN location estimation), Route (computed waypoint sequence with accessibility flag), and CampusDatabase (PostGIS-backed spatial store). Location objects represent physical points in the campus spatial graph and are produced by the positioning subsystem.

Key Class Responsibilities

Class	Responsibility
NavigationSession	**Orchestrates the full navigation lifecycle: receives the query, coordinates NLP + positioning, triggers route computation, and manages session state.**

Class	Responsibility
NLPEngine	**Sends the user query to OpenAI GPT-4 API, parses the structured response, and returns intent + entity pairs for downstream processing.**
WiFiFingerprint	**Collects RSSI signal vectors from visible access points, runs k-NN inference against the fingerprint database, and returns an estimated coordinate pair.**
Route	**Encapsulates the computed waypoint sequence, stores accessibility flags, and generates human-readable turn-by-turn instructions.**
CampusDatabase	**Provides spatial query capabilities via PostGIS; responds to name-based, type-based, and coordinate-proximity-based location queries.**
Location	**Represents a physical campus point with building, floor, room number, and coordinate attributes. Extended by Room, Office, and Facility subtypes.**

3.5.4 Dynamic Models

Dynamic models describe the temporal and behavioral aspects of the system. Two diagram types are presented: a sequence diagram illustrating the navigation query flow, and a state diagram showing the navigation session lifecycle.

Sequence Diagram — Navigation Query Flow

The sequence diagram below traces the complete message exchange for the primary use case: a user submits a navigation query and receives a rendered route. The diagram highlights the asynchronous nature of the NLP API call and the internal route computation step that occurs within the FastAPI server before the result is returned to the client. The validation layer (step 10) ensures that AI-generated location data is cross-checked against the structured campus database before being presented to the user.

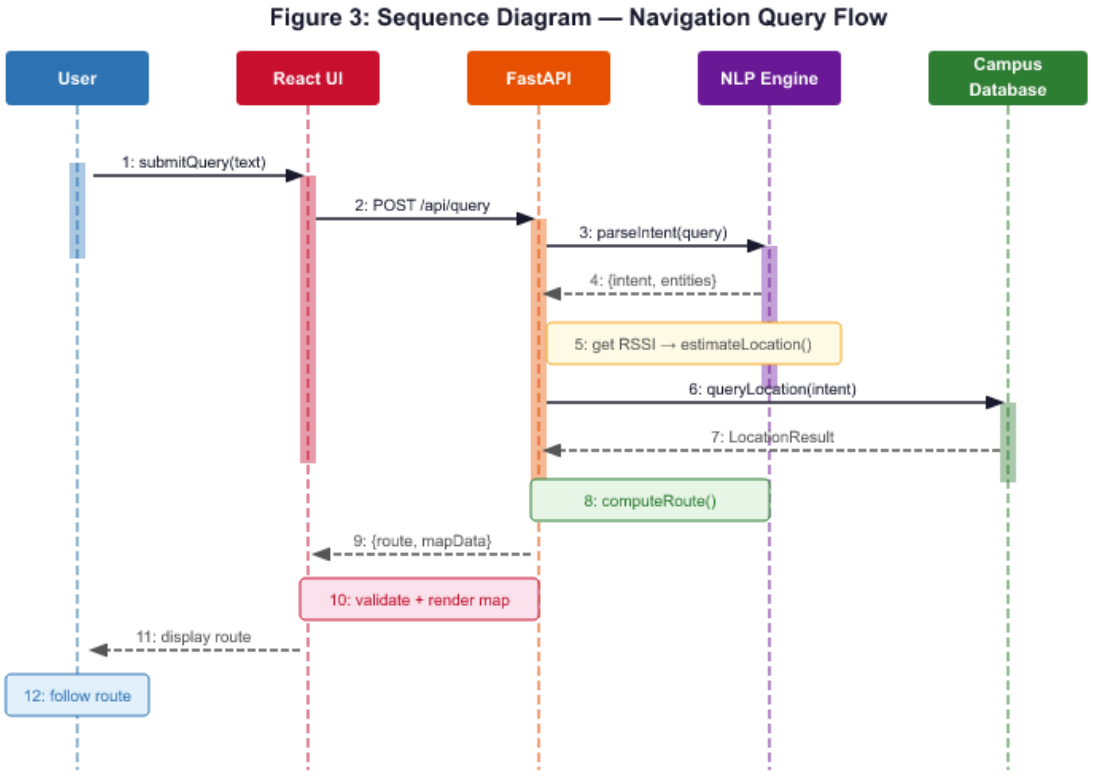

Figure 3: Sequence Diagram — Navigation Query Flow

State Diagram — Navigation Session Lifecycle

The state diagram models the lifecycle of a single navigation session, from the initial IDLE state through PROCESSING, ROUTE GENERATED, and NAVIGATING, to the terminal arrived state. Error paths are shown as dashed transitions, triggering graceful fallback behavior (e.g., manual location selection prompt) when NLP or database operations fail. The 'new query issued' back-edge allows the user to restart navigation mid-journey without terminating the session.

Figure 4: State Diagram — Navigation Session Lifecycle

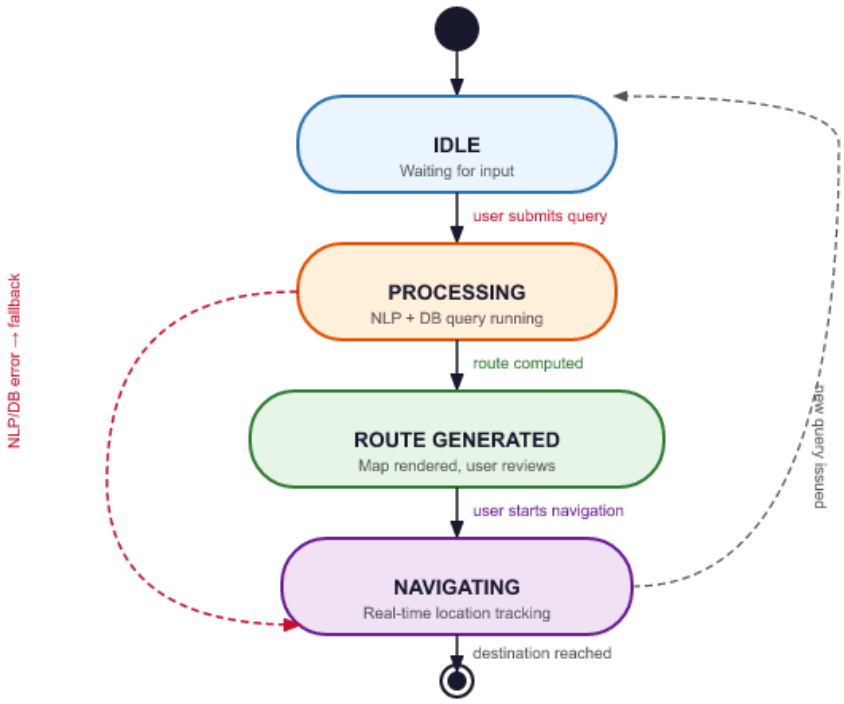

Figure 4: State Diagram — Navigation Session Lifecycle

3.5.5 User Interface — Navigational Paths and Screen Mock-ups

The user interface is organized as a five-screen linear flow, with back-navigation available at all stages. The primary navigational path begins at the Home Screen, proceeds through Search/Results, transitions to the Map View for route visualization, and enters Navigation Mode for turn-by-turn guidance. All screens share a persistent navigation bar providing access to accessibility settings and voice mode.

Figure 5: User Interface — Navigational Paths

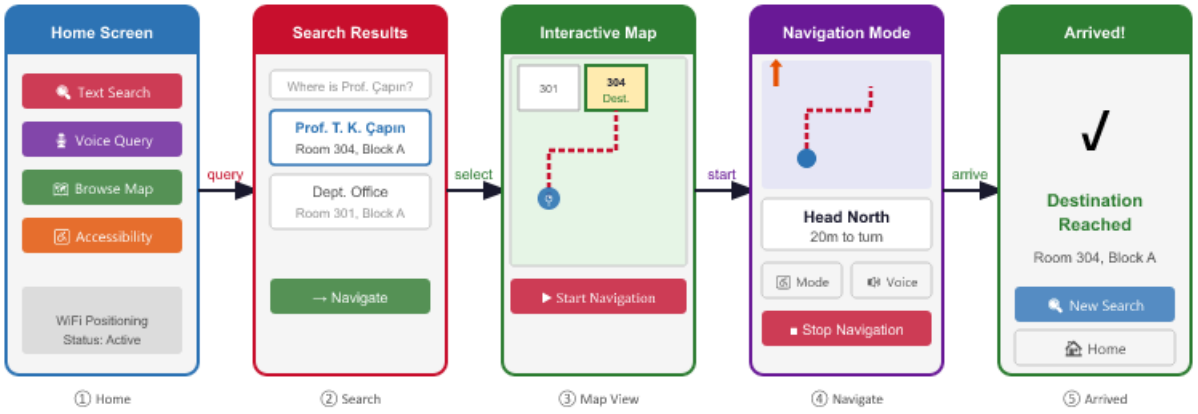

Figure 5: User Interface — Navigational Paths and Screen Mock-ups

Screen Descriptions

Screen 1 — Home Screen: The entry point displays the TEDU Guide brand, a prominent text search field, a voice query trigger button, a map browse shortcut, and a WiFi positioning status indicator. The accessibility toggle is accessible from the home screen.

Screen 2 — Search Results: Displays a ranked list of locations matching the user's query, showing room number, building, and entity type (classroom, office, lab, facility). Each result card includes a 'Navigate' action button. The original query is displayed in the search bar for easy editing.

Screen 3 — Map View: Renders the interactive Leaflet.js floor plan of the relevant building, with the computed route highlighted as a colored overlay. The user's current position is shown as a blue dot. The destination is marked with a pin. Floor switching controls are visible for multi-floor routes.

Screen 4 — Navigation Mode: Activates real-time step-by-step guidance. A large directional arrow indicates the next turn. The remaining distance is displayed numerically. Voice output and haptic feedback toggles are available. A 'Stop Navigation' button terminates the session.

Screen 5 — Arrived Confirmation: Displayed when the user's estimated position matches the destination within the accuracy threshold (3–5 m). Offers options to start a new search or return to the home screen.

3.6 Risk Analysis and Mitigation Strategies

The TEDU Guide system involves several practical risks that may affect its performance in real campus environments. The fluctuation of WiFi RSSI signals brought on by ambient factors, including population density, device interference, and physical barriers, is one of the main risks. The system will use signal smoothing techniques and recalibrate the fingerprints database on a regular basis to lessen this.

The employment of the GPT-4 powered NLP engine, which could produce inaccurate or unclear interpretations of user queries, is another possible concern. By cross-referencing AI-generated outputs with the structured campus database, a validation layer mitigates this risk and guarantees that the user is only presented with verified information.

Another issue is system availability, especially when there is a network outage. The system uses short-term offline support by storing the most recent location and route information in order to lessen the impact.

Lastly, a large user load may result in scalability problems. The system’s stateless API services and cloud-based architecture allow for horizontal scaling when needed. Together, these mitigation techniques guarantee the system’s resilience and dependability in practical situations.

3.7 System Validation and Evaluation Plan

An organized validation strategy will be carried out to assess the TEDU Guide system’s efficacy. By comparing anticipated user locations with ground truth coordinates across several predetermined reference sites within school buildings, the accuracy of indoor positioning will be assessed. Under typical circumstances, the target accuracy is within three to five meters.

In order to verify compliance with the specific performance criterion of less than two seconds, system response time will also be tested under various degrees of concurrent usage. Usability testing sessions including students and visitors will be used to assess the

user experience, with an emphasis on overall happiness, ease of navigation, and clarity of instructions.

Using a predetermined query dataset, the accuracy of intent recognition and response correctness will be used to evaluate the NLP component.

3.8 Error Handling and Edge Cases

The system is built to smoothly manage a variety of error situations. The system will ask the user to manually choose their current location from the map if Wifi connections are not strong enough for precise placement. Alternative search results and fallback suggestions will be displayed if the NLP engine is unable to accurately interpret a query.

Temporary caching of previous loaded map data and routes is used to handle network failures. Incomplete or invalid investigations will also result in user-friendly error messages that advise users to rephrase their requests. Even in less than ideal circumstances, these methods maintain userfulness and guarantee service continuation.

4. Glossary

Term	Definition
RSSI	**Received Signal Strength Indicator — a measurement of the power level of a received radio signal from a WiFi access point, expressed in dBm.**
WiFi Fingerprinting	**An indoor positioning technique that matches a device's observed RSSI vector from multiple access points against a pre-collected database of reference RSSI measurements at known locations.**
k-NN	**k-Nearest Neighbors — a supervised machine learning algorithm used to estimate a user's location by finding the k reference points whose stored RSSI vectors are most similar to the observed vector.**
NLP	**Natural Language Processing — the field of AI concerned with enabling computers to understand, interpret, and generate human language.**
Intent Recognition	**The NLP task of identifying the user's goal from a free-form query (e.g., 'navigate', 'locate', 'query hours').**
Entity Extraction	**The NLP task of identifying specific named objects in a query (e.g., room numbers, person names, facility types).**
PostGIS	**An extension to the PostgreSQL relational database that adds support for storing, indexing, and querying geographic objects and spatial data types.**
Leaflet.js	**An open-source JavaScript library for rendering interactive maps in web browsers, used in TEDU Guide for floor plan display and route overlay.**
FastAPI	**A modern, high-performance Python web framework used to implement the TEDU Guide backend REST API.**
KVKK	**Kişisel Verilerin Korunması Kanunu — Turkish Law No. 6698 on Protection of Personal Data, governing the collection and processing of personal data in Turkey.**

Term	Definition
Accessibility Mode	**An operating mode that restricts navigation routes exclusively to step-free, elevator-accessible paths for users with mobility impairments.**
Haptic Feedback	**Tactile vibration signals produced by a mobile device to convey navigation cues (one pulse = left turn, two pulses = right turn).**
Campus DB	**The structured PostgreSQL/PostGIS database containing all campus location records, including room assignments, faculty offices, facilities, and their coordinates.**
Session	**A bounded interaction period during which a user's navigation query is processed, a route is generated, and location tracking is active.**
Validation Layer	**A backend component that cross-checks AI-generated responses against the campus database before presenting them to the user, preventing hallucinated or incorrect information.**

5. References

1. ACM. (2018). ACM Code of Ethics and Professional Conduct. Association for Computing Machinery. https://www.acm.org/code-of-ethics
2. Aggarwal, P., & Chugh, N. (2022). Review of machine learning techniques for indoor positioning. Journal of Location Based Services, 16(4), 280–322. https://doi.org/10.1080/17489725.2022.2161198
3. Bahl, P., & Padmanabhan, V. N. (2000). RADAR: An in-building RF-based user location and tracking system. Proceedings of IEEE INFOCOM 2000, 2, 775–784. https://doi.org/10.1109/INFCOM.2000.832252
4. Bynum, T. W. (2023). Computer and Information Ethics. Stanford Encyclopedia of Philosophy. https://plato.stanford.edu/entries/ethics-computer/
5. IEEE. (n.d.). IEEE Code of Ethics. Institute of Electrical and Electronics Engineers. https://www.ieee.org/about/corporate/governance/p7-8.html
6. IEEE Computer Society. (1999). IEEE Software Engineering Code of Ethics and Professional Practice (Version 5.2). IEEE/ACM. https://www.computer.org/education/code-of-ethics
7. Liu, H., Darabi, H., Banerjee, P., & Liu, J. (2007). Survey of wireless indoor positioning techniques and systems. IEEE Transactions on Systems, Man, and Cybernetics, Part C, 37(6), 1067–1080. https://doi.org/10.1109/TSMCC.2007.905750
8. OpenAI. (2024). GPT-4 Technical Report. OpenAI. https://openai.com/research/gpt-4
9. Republic of Turkey. (2016). Kişisel Verilerin Korunması Kanunu — Law No. 6698 (Law on Protection of Personal Data). Official Gazette No. 29677.
10. Brusilovsky, P., & Millán, E. (2007). User models for adaptive hypermedia and adaptive educational systems. In The adaptive web: Methods and strategies of web personalization (pp. 3–53). Springer. https://doi.org/10.1007/978-3-540-72079-9_1